Muhammet Celal Gokalp

Technical Application Engineer | Simulation, SIL / HiL & Model-Based Development

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PROFESSIONAL SUMMARY

Customer-facing application engineer with nearly 8 years in safety-critical automotive systems, from ADAS features at Jaguar Land Rover to an L4 automated driving stack at FiveAI / Bosch. Hands-on across the validation lifecycle: requirements, model-based work in MATLAB / Simulink and IPG CarMaker, SIL simulation, and HiL / ViL and vehicle testing. Much of my work is helping engineers adopt new simulation tools and workflows, so I am comfortable explaining technical capability in terms of customer value. I want to move into a technical sales role and bring my automotive validation experience to aerospace customers.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch — Cambridge, United Kingdom

- **Simulation product adoption:** On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Understood users' workflows and pain points, onboarded teams onto the simulation toolchain, and triaged issues across tools, maps and data with several engineering teams.
- **SIL verification at scale:** Built 120+ parameterised OpenSCENARIO 2 scenarios generating thousands of test cases for an L4 driving stack (UN R79 lane change, Euro NCAP junction tests), run locally in a dev container and through the CI test pipeline.
- **Safety-critical evidence:** Defined pass/fail and KPI criteria and analysed results across 1,000+ hours of test data, giving clear findings to engineering and stakeholders.
- **Demonstrating new methods:** Built AI-assisted agent skills (GitHub Copilot) that turn regulation and test protocols into scenarios. I demonstrated them and coached colleagues until they were adopted across the project, increasing scenario output by an order of magnitude.
- **Tools & dashboards:** Built Python tools and data pipelines that present simulation results on dashboards for engineers and decision-makers.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- **Model-based simulation (Simulink / CarMaker):** On a project comparing real vehicle, dyno with sensor emulation from recorded data, and pure simulation testing, calibrated a Simulink and IPG CarMaker vehicle model against real vehicle data, including parameter sweeps on steering torque PID gains.
- **HiL & ViL validation:** Led verification of ADAS features on HiL and ViL rigs and on the proving ground, analysing ECU and network data with Vector CANoe and CANape.
- **Field diagnostics:** At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- **Customer-embedded:** Based at Jaguar Land Rover as feature owner for Rear Cross Traffic Alert, Blind Spot Monitor and Rear Pre-Crash. Main technical link between the OEM, AVL and suppliers, presenting findings and agreeing solutions.
- **Safety-critical process:** Wrote requirements and test cases with traceability in DOORS and MKS Integrity (ASPICE V-model). Wrote ISO 26262 requirements for a lane change feature; trained in ISO 26262 and SOTIF.

Part-Time Teacher and Program Planner

12/2013 – 01/2017

Space Camp Turkey — Izmir, Türkiye

- Taught space technology, physics and science to students aged 9 to 16, explaining complex aerospace topics simply.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation: experimental thermal modelling of an electric motor stator. Coursework included System Modelling and Simulation (MATLAB / Simulink) and Programming (C++).

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: designed, built and controlled a 6-DOF Stewart motion platform, of the kind used in flight simulators (state-space modelling in MATLAB, control on Arduino).

SKILLS

Simulation & Validation	SIL, HiL / ViL, model calibration against real data, scenario-based testing, regression in CI, KPI analysis
Model-Based Tools	MATLAB / Simulink, IPG CarMaker, MotorCAD, OpenSCENARIO 2
Safety-Critical	ISO 26262, SOTIF, ASPICE V-model, requirements traceability (DOORS, MKS Integrity)
Software & Tools	Python, C++ (reading), Git, dev containers, Vector CANoe / CANape, Jira / Confluence, GitHub Copilot
Customer Skills	Workflow discovery, demonstrations, onboarding and training, supplier and OEM liaison. English (C2), Turkish (Native)